Evaluation Metric	Retell AI	Vapi AI	ElevenLabs Conversational AI	Bland AI	Play.ai (PlayHT)	Synthflow AI
1. ENTRY POINTS & SPEED-TO-LEAD						
Inbound Entry Channels	Web Form Webhook	WebRTC Widget / PSTN	React/Web SDK, WebSocket API, PSTN/Twili	API / Inbound Call / QR Code	Web Widget, API Webhooks, Inbound	Webhook, Zapier, Native CRM Forms
Call Speed (Speed-to-Lead)	<10 seconds	<5 seconds	<5 seconds	<15 seconds	<10 seconds	<10 seconds
Multi-Channel Triggers	Parallel SMS / Webhook	Real-time Webhook	Webhook Triggers	Zapier / Custom Webhook	SMS / Webhook Triggers	SMS + Email Sequences
2. TECHNICAL PERFORMANCE & AUDIO						
Latency (Loop Speed)	~600ms total loop	~500ms - 800ms (Modular)	~400ms - 600ms	~700ms total loop	~600ms - 800ms	~700ms - 900ms
Telephony / Transport	WebSockets / Twilio / SIP	WebSockets / WebRTC / PSTN	WebSockets / WebRTC / Twilio	WebSockets / SIP Trunking	WebSockets / SIP / Web Widget	WebSockets / Plivo / Twilio
ASR / STT (Speech-to-Text)	Deepgram Nova-2	Deepgram Nova-2 / Gladia / Talkscriber	Scribe Realtime STT	Proprietary High-Speed ASR	Deepgram Nova-2 / OpenAI Whisper	Custom Fine-Tuned ASR Models
Brain / Reasoning (LLM)	GPT-4o-mini / Claude 3.5 Sonnet	Groq LPU (Llama 3.1) / OpenAI GPT-4o	Native LLM Orchestrator	Pathway LLM Engine	GPT-4o-mini	Workflow Engine LLM
TTS (Text-to-Speech)	ElevenLabs / Cartesia Sonic	Cartesia Sonic / PlayHT / ElevenLabs	Eleven Turbo v2.5 / Flash v2.5	In-House Audio Synthesis ("Bland Turbo")	Play3.0 Mini Conversational Model	Fine-Tuned Voice Models
Orchestration & VAD	Silero VAD (<50ms barge-in) + Dynamic State Machine	Edge VAD (<50ms cut-off) + Modular BYOK Function Calling	Native Low-Latency VAD (<50ms) + Client/Se	Custom VAD + Pathway Node Tree Actions	Standard VAD Thresholds + Convers	Native VAD + No-Code Webhooks & Za
Audio Cleanup	High AEC (Acoustic Echo Cancellation) & Adaptive Noise Filter	Vendor-dependent AEC & Noise Suppression	Built-in Low-Latency Noise & Echo Suppressi	Proprietary Audio Noise Suppression Stack	Standard AEC & Background Noise S	Integrated Noise Cancellation Engine
Barge-in & Interruption	Silero VAD <50ms audio cut-off & buffer clear	Adjustable VAD sensitivity & custom barge-in threshold	Sub-50ms instant audio cut-off VAD	Sensitive VAD cut-off with automatic audio pause	Adjustable VAD Threshold & natural i	Native VAD Interruption & Graceful Re-
3. SCRIPTING, TONE & DYNAMIC STRATEGY						
Hook Strategy	Direct, short question	High-energy conversational	Highly natural, conversational opening	Immediate value offer	Direct value-oriented hook	Workflow-focused prompt hook
Tone & Voice Dynamics	High prosody, natural pauses	Very conversational	Industry-leading prosody & emotion	Controlled, crisp tone	Fluent, warm & expressive tone	Professional, clear business tone
Scripting Approaches	Dynamic Prompt / State Flow	Flexible Dynamic Nodes	Dynamic Agent System Prompts	Pathway Node Tree	Conversational Flow Builder	No-code Visual Workflow Engine
Company & Product Context	Dynamic RAG Lookup	Knowledge Base API	Knowledge Base RAG Docs	Custom Context Inject	Custom Knowledge Uploads	Native Document RAG
Objection Handling	Real-time LLM Rebuttals	Custom Tool Calling	Real-time LLM Tool Calling	Node-based Fallbacks	Prompt-guided Rebuttals	Logic Node Branching
4. CONVERSION MECHANICS & IN-CALL ACTIONS						
Conversion Flow	Lead -> Live Demo Booking	Lead -> Payment / Booking	Lead -> Interactive Booking / Action	Lead -> Human Handoff	Lead -> Demo Slot / Appointment	Lead -> CRM Update / Calendar Booki
In-Call Action Triggers	Live Tool Calling API	Real-time Function Execution	Native Client & Server Tool Calls	Pathway Node Actions	Webhook & Function Execution	Direct API & Zapier Actions
Data Extraction Accuracy	High CRM Field Extraction	High Structured JSON Extraction	High JSON Schema Extraction	Medium Entity Parsing	High Structured Data Extraction	High Field Mapping Accuracy
5. PROVIDER ANALYSIS & COSTING						
Company Positives & Negatives	Pros: Low latency / Cons: Dev setup	Pros: Modular / Cons: 3rd party SLAs	Pros: Highest voice quality / Cons: Higher co	Pros: High scale / Cons: VAD noise	Pros: Fast voice setup / Cons: Webh	Pros: No-code builder / Cons: Less fin
Provider Challenges & Solutions	Packet loss -> Audio buffer tune	Latency sync -> Unified WS bridge	Cost optimization -> Streaming Turbo models	Misfires -> Noise suppression	Turn-taking delays -> VAD tuning	Integration limits -> Native Zapier/Mak
Pricing & Cost Structure	~$0.08 / min = (Telephony: Twilio/SIP) ~$0.015/min + (STT: Deepgram Nova-2) ~$0.006/min + (LLM: GPT-4o-mini) ~$0.015 /min + (TTS: ElevenLabs/Cartesia) ~$0.040 /min + (Platform Fee) ~$0.004 /min	~$0.07 - $0.12 / min = (Telephony: Carrier) ~$0.015/min + (STT: Deepgram/Whisper) ~$0.005/min + (LLM: Groq/OpenAI) ~$0.015/min + (TTS: PlayHT/Cartesia) ~$0.035/min + (Vapi Platform Fee) ~$0.050/min	~$0.09 - $0.15 / min = (Telephony: Twilio/SIP) ~$0.015/min + (STT: Scribe Realtime) ~$0.006/min + (LLM: Reasoning) ~$0.015/min + (TTS: Eleven Turbo v2.5) ~$0.060 - $0.114/min	~$0.09 / min = (Telephony: SIP Trunking) ~$0.015/min + (STT: Proprietary ASR) ~$0.010/min + (LLM: Pathway Engine) ~$0.020/min + (TTS: In-House Audio) ~$0.045 /min	~$0.08 - $0.12 / min = (Telephony: SIP/Plivo) ~$0.015/min + (STT: Deepgram/Whisper) ~$0.006/min + (LLM: GPT-4o-mini) ~$0.015 /min + (TTS: Play3.0 Mini) ~$0.044 - $0.084/min	~$0.09 - $0.13 / min = (Telephony: Plivo/Twilio) ~$0.015/min + (STT: Custom ASR) ~$0.008/min + (LLM: Workflow Engine) ~$0.015/min + (TTS: Fine-Tuned Voice) ~$0.052 - $0.092/min
6. POST-CALL AUTOMATION & WORKFLOWS						

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CRM Writeback & Sync	HubSpot Sync + SMS Invite	Real-time JSON Payload	Webhook -> Custom CRM / API	Webhook -> Zapier -> CRM	Webhook / Zapier Sync	Direct HubSpot & GoHighLevel Sync
Multi-Touch Sequences	Immediate Calendar SMS	Custom API Webhook	Webhook-triggered Follow-up	Automated Follow-up SMS	Automated SMS / Email	Native SMS + Email Workflows
Warm Transfer Success Rate	Live Transfer Supported	Custom SIP Hand-off	Call Transfer Tool Call	Native Human Transfer	SIP Call Transfer	Native Live Call Handoff
7. CALL DELIVERY, COMPLIANCE & FAILURE MODES						
Call Delivery Metrics	~45-50% Answer Rate	Vendor Telephony Dependent	High Quality Transport	High Delivery Rate	~45% Answer Rate	~50% Answer Rate
Call Duration (AHT)	~1 min 30 secs average	Variable by use case	~1 min 30 secs	~1 min 45 secs average	~1 min 40 secs	~1 min 30 secs
Retry & Cadence Logic	Custom Webhook Schedule	API-triggered retries	Custom API Schedule	Dynamic Cadence Rules	Webhook-triggered Retry	Built-in Cadence Scheduler
Compliance & Regulatory	TCPA / ZDR Supported	Vendor Compliance Options	SOC2 / ZDR Supported	TCPA Compliant	TCPA Compliant	TCPA & HIPAA Options
Failure & Edge Case Rate	Low Hallucination Rate	Dependent on LLM selection	Low Voice Artifact Rate	Occasional VAD misfires	Low Error Rate	Fallback Graceful Exit
8. OBSERVATIONS & NOTES ABOUT PROVIDER						
Qualitative Observations	Smooth, human-like flow	Highly flexible UI	Unmatched natural voice prosody & emotion	Fast, enterprise-focused	Extremely fast deployment with good	Easiest no-code platform for agency/S
Strategic Takeaways	Best for custom dev setups	Best for WebRTC/App voice	Best for high-end conversational & brand voic	Best for high-volume cold calls	Best for quick voice agent deployme	Best for no-code CRM-integrated sales
Internal Re-enactment Notes	Copy WebSockets audio queue	Copy unified bridge pattern	Replicate Turbo v2.5 streaming latency patte	Copy noise filter logic	Replicate simplified agent prompt st	Replicate no-code webhook automatic